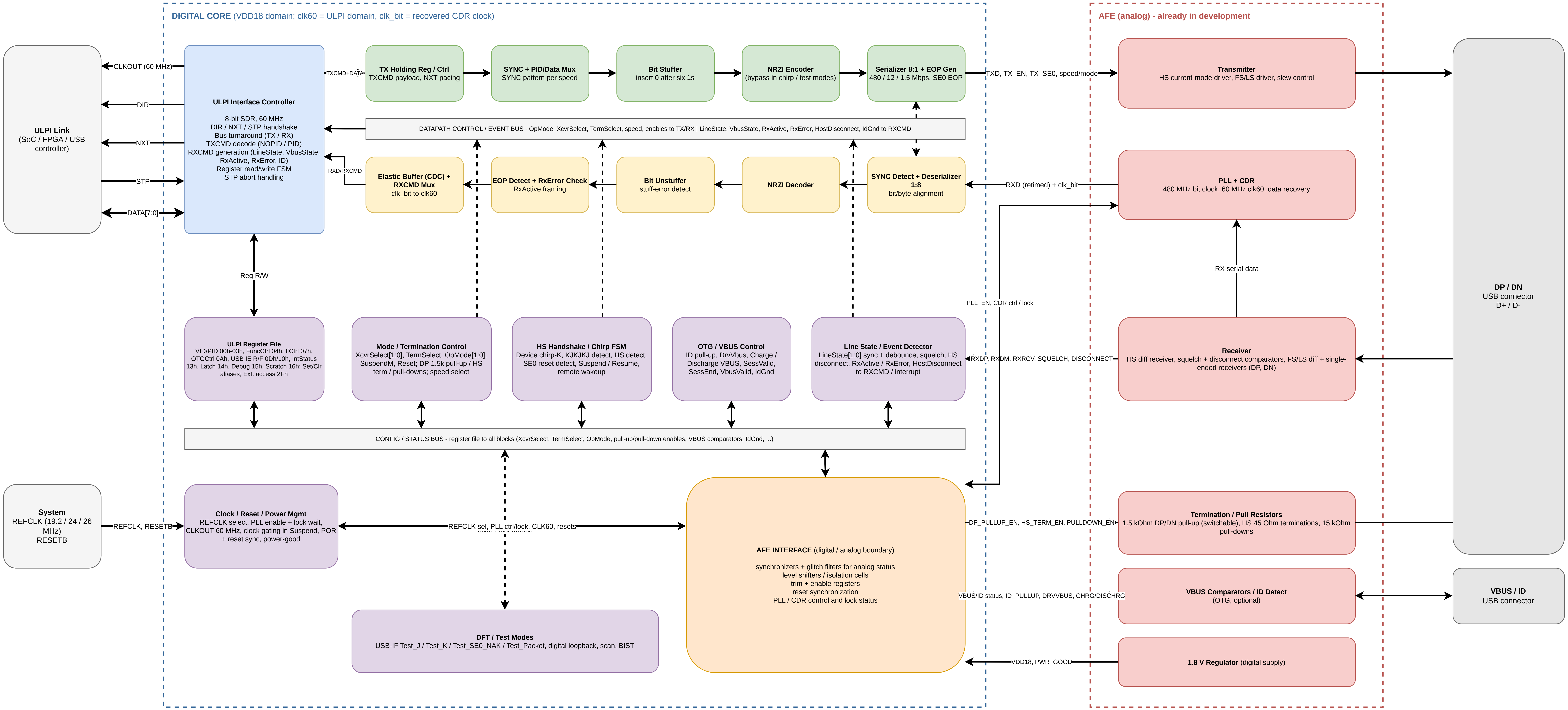
USB 2.0 PHY (HS/FS/LS) - Digital Part with ULPI Interface (reference: USB3320)



DIGITAL CORE (VDD18 domain; clk60 = ULPI domain, clk_bit = recovered CDR clock)
AFE (analog) - already in development
ULPI Link
(SoC / FPGA / USB controller)
CLKOUT (60 MHz)
DIR
NXT
STP
DATA[7:0]
ULPI Interface Controller
8-bit SDR, 60 MHz
DIR / NXT / STP handshake
Bus turnaround (TX / RX)
TXCMD decode (NOPID / PID)
RXCMD generation (LineState, VbusState, RxActive, RxError, ID)
Register read/write FSM
STP abort handling
TXCMD+DATA
TX Holding Reg / Ctrl
TXCMD payload, NXT pacing
SYNC + PID/Data Mux
SYNC pattern per speed
Bit Stuffer
insert 0 after six 1s
NRZI Encoder
(bypass in chirp / test modes)
Serializer 8:1 + EOP Gen
480 / 12 / 1.5 Mbps, SE0 EOP
TXD, TX_EN, TX_SE0, speed/mode
Transmitter
HS current-mode driver, FS/LS driver, slew control
DATAPATH CONTROL / EVENT BUS - OpMode, XcvrSelect, TermSelect, speed, enables to TX/RX | LineState, VbusState, RxActive, RxError, HostDisconnect, IdGnd to RXCMD
RXD/RXCMD
Elastic Buffer (CDC) + RXCMD Mux
clk_bit to clk60
EOP Detect + RxError Check
RxActive framing
Bit Unstuffer
stuff-error detect
NRZI Decoder
SYNC Detect + Deserializer 1:8
bit/byte alignment
RXD (retimed) + clk_bit
PLL + CDR
480 MHz bit clock, 60 MHz clk60, data recovery
RX serial data
PLL_EN, CDR ctrl / lock
DP / DN
USB connector
D+ / D-
Reg R/W
ULPI Register File
VID/PID 00h-03h, FuncCtrl 04h, IfCtrl 07h, OTGCtrl 0Ah, USB IE R/F 0Dh/10h, IntStatus 13h, Latch 14h, Debug 15h, Scratch 16h; Set/Clr aliases; Ext. access 2Fh
Mode / Termination Control
XcvrSelect[1:0], TermSelect, OpMode[1:0], SuspendM, Reset; DP 1.5k pull-up / HS term / pull-downs; speed select
HS Handshake / Chirp FSM
Device chirp-K, KJKJKJ detect, HS detect, SE0 reset detect, Suspend / Resume, remote wakeup
OTG / VBUS Control
ID pull-up, DrvVbus, Charge / Discharge VBUS, SessValid, SessEnd, VbusValid, IdGnd
Line State / Event Detector
LineState[1:0] sync + debounce, squelch, HS disconnect, RxActive / RxError, HostDisconnect to RXCMD / interrupt
RXDP, RXDM, RXRCV, SQUELCH, DISCONNECT
Receiver
HS diff receiver, squelch + disconnect comparators, FS/LS diff + single-ended receivers (DP, DN)
CONFIG / STATUS BUS - register file to all blocks (XcvrSelect, TermSelect, OpMode, pull-up/pull-down enables, VBUS comparators, IdGnd, ...)
System
REFCLK (19.2 / 24 / 26 MHz)
RESETB
REFCLK, RESETB
Clock / Reset / Power Mgmt
REFCLK select, PLL enable + lock wait, CLKOUT 60 MHz, clock gating in Suspend, POR + reset sync, power-good
REFCLK sel, PLL ctrl/lock, CLK60, resets
scan / Test modes
AFE INTERFACE (digital / analog boundary)
synchronizers + glitch filters for analog status
level shifters / isolation cells
trim + enable registers
reset synchronization
PLL / CDR control and lock status
DP_PULLUP_EN, HS_TERM_EN, PULLDOWN_EN
Termination / Pull Resistors
1.5 kOhm DP/DN pull-up (switchable), HS 45 Ohm terminations, 15 kOhm pull-downs
VBUS/ID status, ID_PULLUP, DRVVBUS, CHRG/DISCHRG
VBUS Comparators / ID Detect
(OTG, optional)
VBUS / ID
USB connector
DFT / Test Modes
USB-IF Test_J / Test_K / Test_SE0_NAK / Test_Packet, digital loopback, scan, BIST
VDD18, PWR_GOOD
1.8 V Regulator (digital supply)

Clock domains: clk60 = ULPI / digital domain (60 MHz from PLL, output as CLKOUT). clk_bit = clock recovered by CDR (480 / 12 / 1.5 Mbps); the Elastic Buffer crosses RX data into clk60. Dashed red = analog AFE, dashed blue = digital core. Green = TX path, yellow = RX path, purple = control, grey = buses.